# Practice Midterm

Last Name: _____

First Name: _____

Student ID: _____

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**Instructions:** This practice midterm is a bit longer than the real midterm, so that you have more problems to practice on.

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Please do not write below this line.

| Question | Points | Score |
|---|---|---|
| 1 | 20 | |
| 2 | 15 | |
| 3 | 10 | |
| 4 | 15 | |
| 5 | 18 | |
| 6 | 0 | |
| 7 | 12 | |
| Total: | 90 | |

1. Consider the tensor $\mathcal{A} \in \mathbb{R}^{2\times 2\times 3}$ and the matrix $B \in \mathbb{R}^{2\times 2}$ where:

$$\mathcal{A}(:,:,1) = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \quad \mathcal{A}(:,:,2) = \begin{bmatrix} -4 & 0 \\ 3 & 1 \end{bmatrix} \quad \mathcal{A}(:,:,3) = \begin{bmatrix} 2 & -7 \\ 1 & 5 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 5 \\ -2 & 1 \end{bmatrix}$$

(a) (5 points) Can we compute $\mathcal{A} \times_1 B$? If yes, compute this. If no, explain why not.

**Solution:** Yes, as the dimension in mode 1 of $\mathcal{A}$ is equal to the dimension in mode 2 of $B$.

$$(\mathcal{A} \times_1 B)(:,:,1) = \begin{bmatrix} 10 & 20 \\ 0 & -2 \end{bmatrix} \quad (\mathcal{A} \times_1 B)(:,:,2) = \begin{bmatrix} 15 & 5 \\ 11 & 1 \end{bmatrix}$$

$$\text{and } (\mathcal{A} \times_1 B)(:,:,3) = \begin{bmatrix} 5 & 25 \\ -3 & 19 \end{bmatrix}$$

(b) (5 points) Can we compute $\mathcal{A} \times_3 B$? If yes, compute this. If no, explain why not.

**Solution:** No, as the dimension of mode 3 in $\mathcal{A}$ (*i.e.* 3) is not equal to the dimension of mode 2 in $B$ (*i.e.* 2).

(c) (5 points) Explain, in words, what it means to unfold a tensor.

**Solution:** It means to convert a tensor into a matrix. This is done by stacking the *fibers* of $\mathcal{A}$. (Fibers are the 1-dimensional sub-arrays, e.g. rows, columns etc).

(d) (5 points) Give, in words, one application in data science where tensors might be used.

**Solution:** $n \times n$ pixel colour images are frequently stored and processed as $n \times n \times 3$ tensors which have one $n \times n$ slice for each of red green and blue.

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2. Consider the matrix $A \in \mathbb{R}^{3\times 4}$, given below together with its singular value decomposition:

$$A = \begin{bmatrix} 0.5377 & 0.8622 & -0.4336 & 2.7694 \\ 1.8339 & 0.3188 & 0.3426 & -1.3499 \\ -2.2588 & -1.3077 & 3.5784 & 3.0349 \end{bmatrix}$$
$$= \begin{bmatrix} 0.2137 & -0.9463 & -0.2427 \\ -0.2814 & 0.1782 & -0.9429 \\ 0.9355 & 0.2698 & -0.2282 \end{bmatrix} \begin{bmatrix} 5.622 & 0 & 0 & 0 \\ 0 & 2.8458 & 0 & 0 \\ 0 & 0 & 1.7117 & 0 \end{bmatrix} \times$$
$$\begin{bmatrix} -0.4438 & -0.2781 & -0.7852 & -0.3303 \\ -0.1992 & -0.3907 & -0.1235 & 0.8902 \\ 0.5575 & 0.5049 & -0.6044 & 0.2626 \\ 0.6727 & -0.7177 & -0.0537 & -0.1719 \end{bmatrix}$$

(a) (5 points) Give the definition of the matrix 2-norm, $\|A\|_2$.

**Solution:** $\|A\|_2 = \sup_{\substack{\mathbf{x}\in\mathbb{R}^d \\ \mathbf{x}\neq 0}} \frac{\|A\,\mathbf{x}\,\|_2}{\|\,\mathbf{x}\,\|_2} = \max_{\substack{\mathbf{x}\in\mathbb{R}^d \\ \|\,\mathbf{x}\,\|_2=1}} \|A\,\mathbf{x}\,\|_2$ *either definition is equally acceptable*

(b) (5 points) Using the information given, compute $\|A\|_2$.

**Solution:** We know that $\|A\|_2 = \sigma_{\max}(A)$ = largest eigenvalue of $A$. Hence, from the SVD given: $\|A\|_2 = 5.622$.

(c) Compute the matrix $\infty$-norm, $\|A\|_\infty$.

**Solution:** We know that $\|A\|_\infty =$ largest absolute row sum $= \max_j \sum_{i=1}^{n} |A_{ij}|$. so, compute the absolute row sums of $A$ and observe that the third row gives the largest value, namely 10.1798. Hence $\|A\|_\infty = 10.1798$. *Note: This question does not use the SVD in any way.*

(d) (5 points) Compute, using the information given, the best rank 2 approximation to $A$ .

Question 2 continues on the next page...

Question 2 continued...

**Solution:** Suppose that $A = U\Sigma V^\top$ is the SVD of $A$. Recall that the best rank 2 approximation, $A_2$, is given in outer product form as $A = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^\top + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^\top$. From the SVD given, we get that:

$$
\begin{aligned}
A_2 &= 5.622 \begin{bmatrix} 0.2137 \\ -0.2814 \\ 0.9355 \end{bmatrix} \begin{bmatrix} -0.4438 & -0.2781 & -0.7852 & -0.3303 \end{bmatrix} \\
&\quad + 2.8458 \begin{bmatrix} -0.9463 \\ 0.1782 \\ 0.2698 \end{bmatrix} \begin{bmatrix} -0.1992 & -0.3907 & -0.1235 & 0.8902 \end{bmatrix} \\
&= 5.622 \begin{bmatrix} -0.0948 & -0.0594 & -0.1678 & -0.0706 \\ 0.1249 & 0.0783 & 0.2210 & 0.0929 \\ -0.4152 & -0.2602 & -0.7346 & -0.3090 \end{bmatrix} \\
&\quad + 2.8458 \begin{bmatrix} 0.1885 & 0.3697 & 0.1169 & -0.8424 \\ -0.0355 & -0.0696 & -0.0220 & 0.1586 \\ -0.0537 & -0.1054 & -0.0333 & 0.2402 \end{bmatrix} \\
&= \begin{bmatrix} 0.0033 & 0.7180 & -0.6108 & -2.7941 \\ 0.6011 & 0.2418 & 1.1796 & 0.9740 \\ -2.4871 & -1.7626 & -4.2245 & -1.0537 \end{bmatrix}
\end{aligned}
$$

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3. Consider the least squares problem: $\min_{\mathbf{x}\in\mathbb{R}^n} \|A\,\mathbf{x}-\mathbf{b}\,\|_2$ where $A \in \mathbb{R}^{m\times n}$.

   (a) (6 points) What is the QR decomposition of $A$? How would one use this to solve this least squares problem?

**Solution:** The QR decomposition is a decomposition/factorization of $A$ as $A = QR$, where $Q \in \mathbb{R}^{m\times m}$ is orthogonal and $R \in \mathbb{R}^{m\times n}$ can be written as $R = \begin{bmatrix} R_1 \\ 0_{(m-n)\times n} \end{bmatrix}$ where $R_1 \in \mathbb{R}^{n\times n}$ is upper triangular and $0_{(m-n)\times n}$ represents the all-zeros matrix. *Note: this is assuming $m > n$.*

We use it in the following manner:

$$\begin{aligned} \|A\,\mathbf{x}-\mathbf{b}\,\|_2 &= \|QR\,\mathbf{x}-\mathbf{b}\,\|_2 \\ &= \left\|Q^\top (QR\,\mathbf{x}-\mathbf{b})\right\|_2 \text{ (by unitary invariance)} \\ &= \left\|R\,\mathbf{x}-Q^\top\,\mathbf{b}\right\|_2 \text{ because } Q^\top = Q^{-1} \end{aligned}$$

Now rewrite: $\mathbf{x} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix}$ and $Q^\top\,\mathbf{b} = \begin{bmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \end{bmatrix}$ where $\mathbf{x}_1$ and $\mathbf{b}_1$ denote the first $n$ entries of $\mathbf{x}$ and $\mathbf{b}$ respectively. Observe that:

$$\left\|R\,\mathbf{x}-Q^\top\,\mathbf{b}\right\|_2^2 = \|R_1\,\mathbf{x}_1-\mathbf{b}_1\,\|_2^2 + \|0\,\mathbf{x}_2-\mathbf{b}_2\,\|_2^2 = \|R_1\,\mathbf{x}_1-\mathbf{b}_1\,\|_2^2 + \|\,\mathbf{b}_2\,\|_2^2$$

Hence one can minimize $\left\|R\,\mathbf{x}-Q^\top\,\mathbf{b}\right\|_2^2$ (which is equivalent to minimizing $\left\|R\,\mathbf{x}-Q^\top\,\mathbf{b}\right\|_2$) by solving $R_1\,\mathbf{x}_1 = \mathbf{b}_1$. This is easy to do because $R_1$ is upper triangular. *A less detailed answer would also earn full credit.*

   (b) (4 points) Recall that $\kappa(A)$ denotes the condition number of $A$. Which situation would be preferable: $\kappa(A) = 5$ or $\kappa(A) = 10^4$? Justify your answer.

**Solution:** A smaller condition number is always preferable. This is because for any small perturbation in $A$, *e.g.* $A \to A + \delta A$, the difference in the solutions $\mathbf{x}_0 = \text{argmin}\|A\,\mathbf{x}-\mathbf{b}\,\|_2$ and $\mathbf{x}_1 = \text{argmin}\|(A+\delta A)\,\mathbf{x}-\mathbf{b}\,\|_2$ is proportional to $(\kappa(A))^2$. Thus, a smaller condition number means that the least squares problem is less sensitive to errors.

4. For the following statements, state whether they are true or false. If false, explain what is wrong with the statement or provide a corrected version.

   (a) (5 points) For every matrix $A \in \mathbb{R}^{n\times n}$ there exists an upper triangular matrix $U$ and a lower triangular matrix $L$ with ones on its diagonal such that $A = LU$.

   **Solution:** False. You may have to permute the rows. A true statement is:
   *For every matrix $A \in \mathbb{R}^{n\times n}$ there exists an upper triangular matrix $U$,a lower triangular matrix $L$ with ones on its diagonal* **and a permutation matrix** $P$ *such that $PA = LU$*

   (b) (5 points) For every matrix $B \in \mathbb{R}^{m\times n}$ there exists an orthogonal matrix $Q \in \mathbb{R}^{m\times m}$ and a matrix $R \in \mathbb{R}^{m\times n}$ with $R = \begin{bmatrix} R_1 \\ 0 \end{bmatrix}$ where $R_1 \in \mathbb{R}^{n\times n}$ is upper triangular such that $B = QR$.

   **Solution:** True.

   (c) (5 points) For every non-negative matrix $X \in \mathbb{R}^{m\times n}$ and every $r < \min\{m, n\}$ there exists non-negative matrices $H \in \mathbb{R}^{m\times r}$ and $W \in \mathbb{R}^{r\times n}$ such that $X = WH$.

   **Solution:** False. We have no guarantee that a non-negative matrix factorization exists. A true, but vague, statement is the following:
   *For every non-negative matrix $X \in \mathbb{R}^{m\times n}$ and every $r < \min\{m, n\}$ one can find non-negative matrices $H \in \mathbb{R}^{m\times r}$ and $W \in \mathbb{R}^{r\times n}$ such that $X \approx WH$*

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5. (a) (8 points) Write pseudocode for the $k$-means algorithm.

**Solution:**

- Randomly assign the data points $\mathbf{x}_1, \ldots, \mathbf{x}_n$ to $k$ different sets $\mathcal{X}_1, \ldots, \mathcal{X}_k$.
- Until stopping criterion reached:
  1. For $a = 1, \ldots, k$ compute centroid $\boldsymbol{\mu}_a = \frac{1}{n_a} \sum_{\substack{i \\ \mathbf{x}_i \in \mathcal{X}_a}} \mathbf{x}_i$ where $n_a = |\mathcal{X}_a|$.
  2. For $i = 1, \ldots, n$ assign $\mathbf{x}_i$ to $\mathcal{X}_{a_i^*}^{\text{new}}$ where $a_i^* = \text{argmin}_a \| \mathbf{x}_i - \boldsymbol{\mu}_a \|_2$.
  3. Replace the $\mathcal{X}_a$ with $\mathcal{X}_a^{\text{new}}$.

The stopping criterion can either be a maximum number of iterations, or, one can stop when $\mathcal{X}_a = \mathcal{X}_a^{\text{new}}$ for all $a$.

(b) (5 points) Draw one data set, consisting of two clusters, for which $k$-means will correctly find these clusters.

**Solution:** This should be something like the data set from Lecture 7.

(c) (5 points) Now, draw one data set, again consisting of two clusters, for which $k$-means **will not** find these clusters.

**Solution:** This should be something like concentric circles.

6. Consider the following text processing problem. Suppose that we have four documents, and we select 5 key words. The matrix $X$, given below, shows the number of times each keyword appears in each document. For example, $X_{23} = 1$ means that keyword 2 appears once in document 3.

$$X = \begin{bmatrix} 3 & 0 & 1 & 2 \\ 5 & 1 & 1 & 4 \\ 0 & 7 & 2 & 1 \\ 2 & 5 & 1 & 0 \\ 1 & 4 & 1 & 4 \end{bmatrix}$$

Use the tf-idf scheme to make a new, weighted, term matrix $\tilde{X}$. You do not need to normalize the columns.

**Solution:** Let $\mathrm{df}(w_i)$ be the number of documents the $i$-th keyword appears in. We compute these:

$$\mathrm{df}(w_1) = 3 \quad \mathrm{df}(w_2) = 4 \quad \mathrm{df}(w_3) = 3 \quad \mathrm{df}(w_4) = 3 \quad \mathrm{df}(w_5) = 4$$

Here, the tf-idf formula is: $\tilde{X}_{ij} = X_{ij} \log\left(\dfrac{4}{\mathrm{df}(w_i)}\right)$ *(Note that I'm using log base 10 here)*, so we get the new matrix:

$$\tilde{X} = \begin{bmatrix} 0.3748 & 0 & 0.1249 & 0.2499 \\ 0 & 0 & 0 & 0 \\ 0 & 0.8746 & 0.2499 & 0.1249 \\ 0.2499 & 0.6247 & 0.1249 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

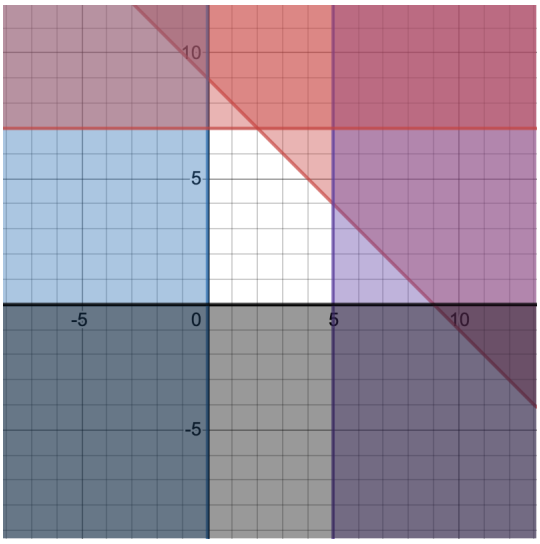
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7. (12 points) Consider the linear program:

$$\begin{aligned} \text{minimize } & 2x_1 + x_2 \\ \text{subject to: } & x_1 + x_2 \leq 9 \\ & x_1 \leq 5 \text{ and } x_2 \leq 7 \\ & x_1 \geq 0 \text{ and } x_2 \geq 0 \end{aligned}$$

Sketch the feasible region and find a solution to this problem.

**Solution:**




Note that the feasible region is the **unshaded** (*i.e.* white) area. From this picture, we deduce that the corners (also known as basic feasible solutions) are $(0, 7), (2.7), (5, 4), (5, 0)$ and $(0, 0)$. One can easily check that $(0, 0)$ gives the minimum value of the cost function among all the corners, so this is the solution to the problem.